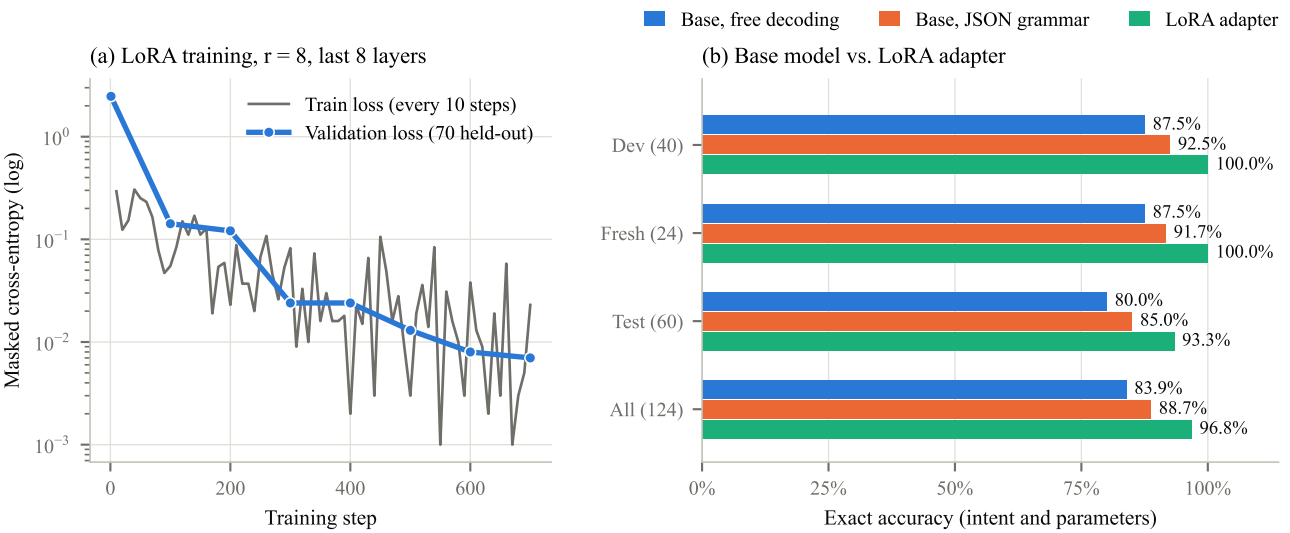
(a) LoRA training, r = 8, last 8 layers
Train loss (every 10 steps)
Validation loss (70 held-out)
Masked cross-entropy (log)
10^{0}
10^{-1}
10^{-2}
10^{-3}
0
200
400
600
Training step
Base, free decoding
Base, JSON grammar
LoRA adapter
(b) Base model vs. LoRA adapter
Dev (40)
87.5%
92.5%
100.0%
Fresh (24)
87.5%
91.7%
100.0%
Test (60)
80.0%
85.0%
93.3%
All (124)
83.9%
88.7%
96.8%
0%
25%
50%
75%
100%
Exact accuracy (intent and parameters)